

Worksheet — Patterns in Mathematics (Class 6 CBSE)

Total marks: 25 · Questions: 11

Instructions: Attempt all 10 questions. Show your working for the subjective questions; choose the single best option for MCQs. Total time: 30 minutes.

Q1. The first five SQUARE numbers are: [1 mark]

- (A) 1, 2, 3, 4, 5
- (B) 1, 4, 9, 16, 25
- (C) 1, 3, 6, 10, 15
- (D) 1, 8, 27, 64, 125

Q2. Two consecutive triangular numbers always add up to a square number. [1 mark]

- (A) True
- (B) False

Q3. The 5th square number is _____. [1 mark]

Q4. What comes next: 1, 3, 6, 10, _____? [1 mark]

- (A) 13
- (B) 14
- (C) 15
- (D) 16

Q5. What is $1 + 3 + 5 + 7 + 9 + 11$ (sum of the first 6 odd numbers)? [2 marks]

- (A) 30
- (B) 32
- (C) 36
- (D) 40

Q6. Find the next two terms of the Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, _____, _____. [2 marks]

Q7. Find the 7th triangular number using the formula $\text{triangular}(n) = n(n+1)/2$. [2 marks]

Q8. Verify that $\text{triangular}(5) + \text{triangular}(6)$ is a square number, and identify which. [3 marks]

Q9. Use the formula $1 + 3 + 5 + \dots + (2n - 1) = n^2$ to find the sum of the first 12 odd numbers. [3 marks]

Q10. Draw (in words) a 5×5 grid of dots. Mark off L-shaped strips of 1 dot, then 3, then 5, then 7, then 9. Why does each strip have an odd number of dots? What is the running total? What does this prove? [5 marks]

Q11. Make up your own sequence with a clear rule. Write the first 6 terms and state the rule. Then say which 'famous' sequence (counting/even/odd/square/triangular/cube/Fibonacci/powers) is most similar to yours and why. [4 marks]

Answer Key

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Q1. The first five SQUARE numbers are:

Answer: 1, 4, 9, 16, 25

Square numbers come from $n \times n$.

MCQ · EASY · 6-CBSE-MATH-PAT-02

Q2. Two consecutive triangular numbers always add up to a square number.

Answer: True

$\text{triangular}(n) + \text{triangular}(n+1) = (n+1)^2$.

TRUE_FALSE · MEDIUM · 6-CBSE-MATH-PAT-06

Q3. The 5th square number is ____.

Answer: 25

$5 \times 5 = 25$.

FILL_BLANK · EASY · 6-CBSE-MATH-PAT-02

Q4. What comes next: 1, 3, 6, 10, ____?

Answer: 15

Triangular numbers; $10 + 5 = 15$.

MCQ · EASY · 6-CBSE-MATH-PAT-03

Q5. What is $1 + 3 + 5 + 7 + 9 + 11$ (sum of the first 6 odd numbers)?

Answer: 36

Sum of first n odd numbers $= n^2 = 6^2 = 36$.

MCQ · MEDIUM · 6-CBSE-MATH-PAT-05

Q6. Find the next two terms of the Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, ____, ____.

Answer: 21, 34. Each term is the sum of the previous two: $8 + 13 = 21$; $13 + 21 = 34$.

$F(n) = F(n-1) + F(n-2)$.

SHORT_ANSWER · MEDIUM · 6-CBSE-MATH-PAT-03

Q7. Find the 7th triangular number using the formula $\text{triangular}(n) = n(n+1)/2$.

Answer: $\text{Triangular}(7) = 7 \times 8 / 2 = 28$.

Plug $n = 7$ into $n(n+1)/2$.

SHORT_ANSWER · MEDIUM · 6-CBSE-MATH-PAT-05

Q8. Verify that $\text{triangular}(5) + \text{triangular}(6)$ is a square number, and identify which.

Answer: $\text{Triangular}(5) = 15$. $\text{Triangular}(6) = 21$. $\text{Sum} = 36 = 6^2$ (the 6th square number).

Demonstrates the consecutive-triangulars-sum-to-square pattern.

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Q9. Use the formula $1 + 3 + 5 + \dots + (2n - 1) = n^2$ to find the sum of the first 12 odd numbers.

Answer: $\text{Sum} = 12^2 = 144$.

Apply the formula directly.

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Q10. Draw (in words) a 5×5 grid of dots. Mark off L-shaped strips of 1 dot, then 3, then 5, then 7, then 9. Why does each strip have an odd number of dots? What is the running total? What does this prove?

Answer: Strip 1 (top-left corner) = 1 dot. Strip 2 (L-shape around it) = 3 dots. Strip 3 = 5 dots. Strip 4 = 7. Strip 5 = 9. Each new strip wraps around the previous square, adding $2n + 1$ dots (always odd). Running totals: 1, 4, 9, 16, 25 — square numbers. So $1 + 3 + 5 + 7 + 9 = 25 = 5^2$. The picture proves that the sum of the first n odd numbers is always n^2 .

Classic visual proof of $1+3+5+\dots+(2n-1) = n^2$.

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Q11. Make up your own sequence with a clear rule. Write the first 6 terms and state the rule.

Then say which 'famous' sequence

(counting/even/odd/square/triangular/cube/Fibonacci/powers) is most similar to yours and why.

Answer: Example: rule 'add 4 each time, starting at 2'. First 6 terms: 2, 6, 10, 14, 18, 22. Most similar to 'multiples of 2 (even numbers)' — both add a constant each time, but mine adds 4 not 2. (Many correct answers; rubric: clear rule + 6 correct terms + comparison to a named sequence.)

Encourages pattern-creation and connection to named sequences.

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